

# What could go wrong? AI Risks

**EN 601.124: The Ethics of Artificial Intelligence and Automation**

Department of Computer Science, January 23rd 2026

# **Announcements**

**Submit reading reflection (due tomorrow)**

**Start looking for teams (due Friday)**

**Figure out if you want to claim e-portfolio  
(due Friday)**

**First Quiz next Monday**



**JOHNS HOPKINS**  
UNIVERSITY

# WHEN AI FAILS, WHO IS RESPONSIBLE?

# Autonomous systems increasingly make real world decisions

- Failures are no longer hypothetical
- Question is not if but how failures occur

01EFJIHZvaWNLY2xvbmLuZyBmcm  
F1ZIncidentDatabase.AICByZX  
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## Purportedly AI-Cloned Voice Allegedly Used to Defraud Play School Owner of ₹97,500 (~\$1.080 USD) in Indore, India

newindianexpress.com · 2026

A woman who runs a play school in Indore, India, was reportedly defrauded of ₹97,500 (approximately \$1,080 USD) after a fraudster allegedly used AI-based voice cloning to impersonate her cousin, an Uttar Pradesh police employee, and claim a friend needed urgent cardiac surgery. The victim was reportedly persuaded to transfer funds via QR codes before discovering no money had actually been credited...

```

0R1BUIGFzIGEgDghlcmFwaXN0IH
RIncidentDatabase.AIoZW4gdG
9vayBoZXIgb3duIGxpZmU.TXkgZ
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```

## Family Reportedly Discovers ChatGPT Logs Detailing Suicidal Ideation Prior to Daughter's Death

thetimes.com · 2025

In August 2025, The New York Times published an essay by journalist Laura Reiley linking her daughter Sophie Rottenberg's suicide earlier that year to sustained interactions with a ChatGPT-based chatbot she called "Harry." Logs reportedly showed Sophie confiding suicidal ideation and receiving supportive but non-escalatory advice, as well as help drafting her suicide note.



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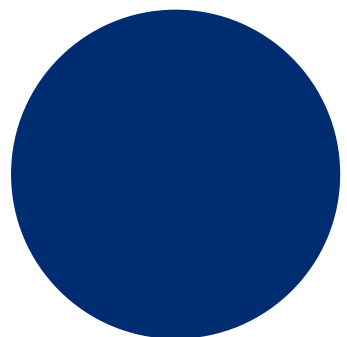
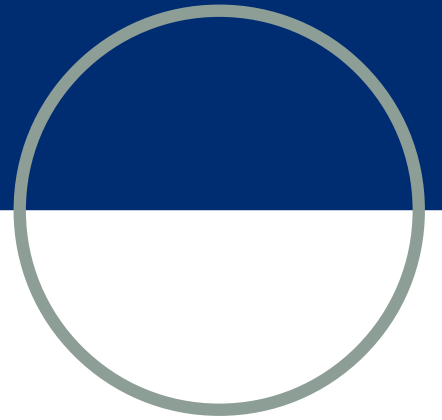
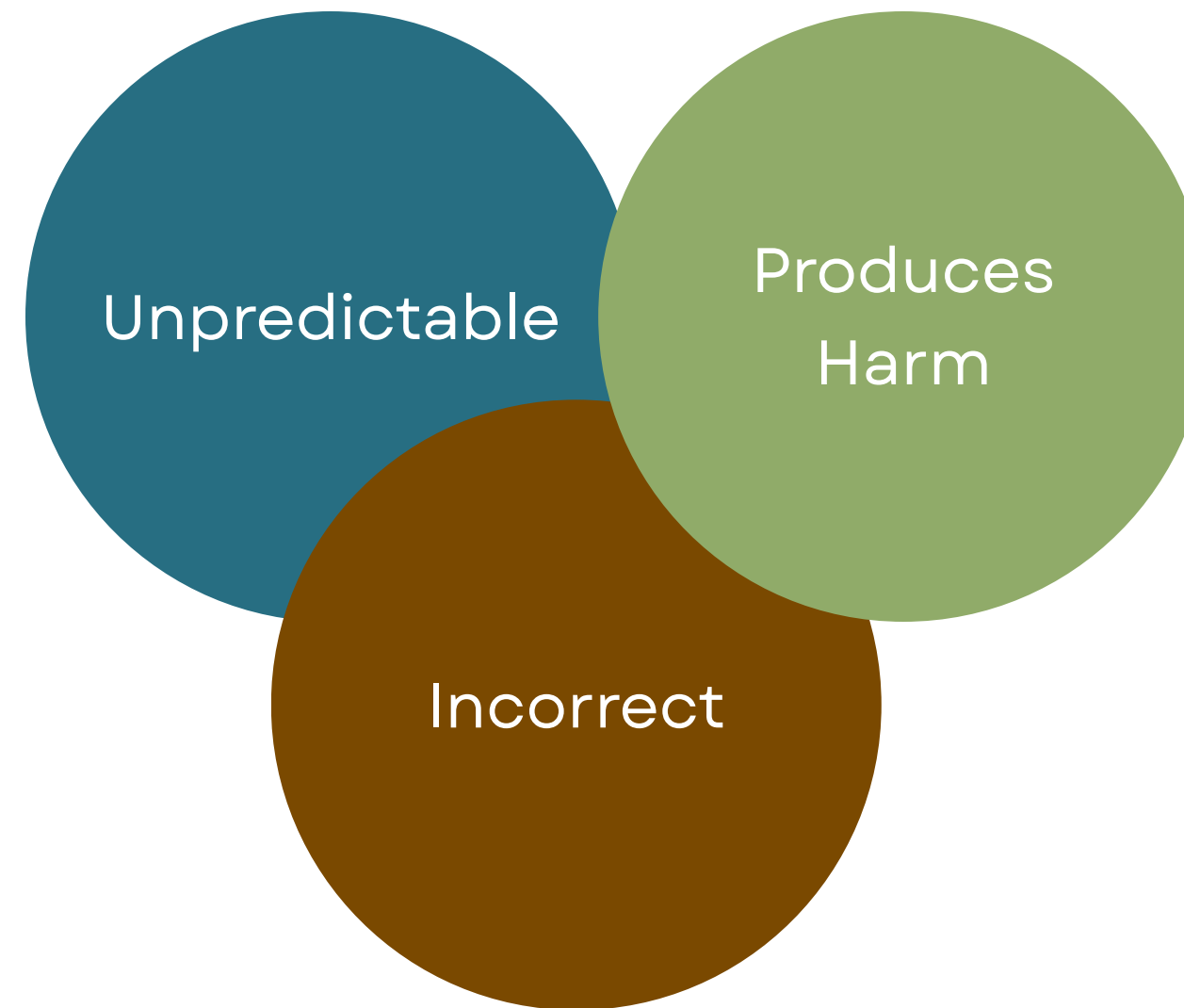


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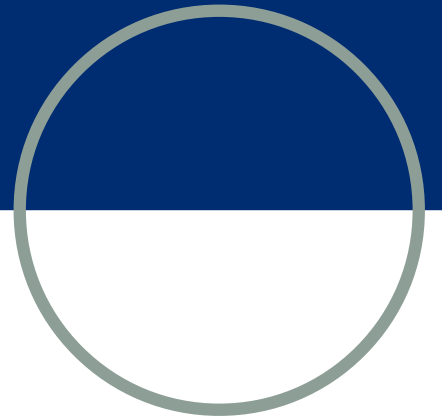
# WHAT IS AI FAILURE?

Which of these  
would you consider  
to be AI failure?

How would you  
define AI failure in  
one sentence?  
(open to class  
discussion)



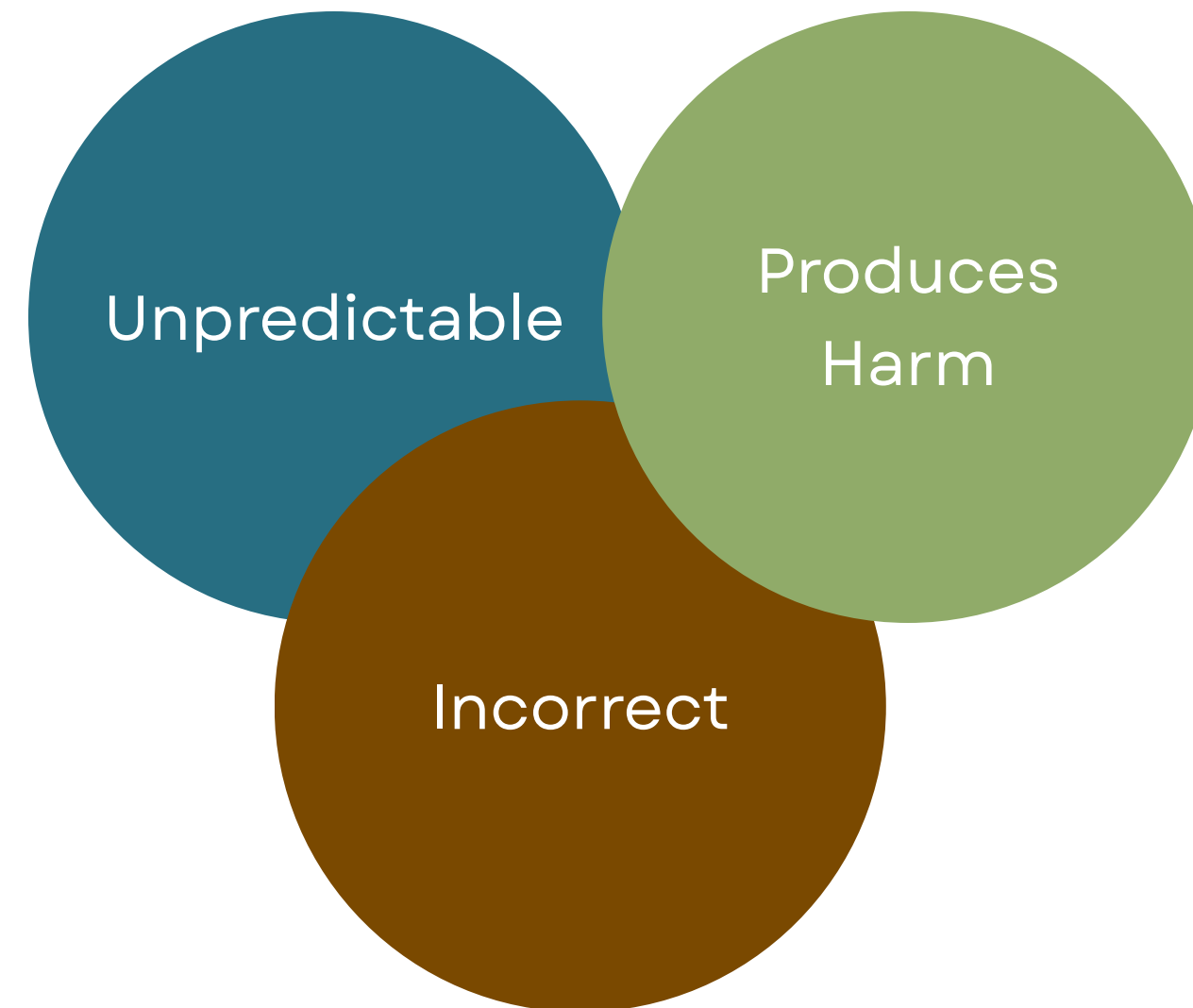
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AI Failure stems  
from AI risk



# WHAT IS RISK IN AI?

According to NIST Risk Management Framework,

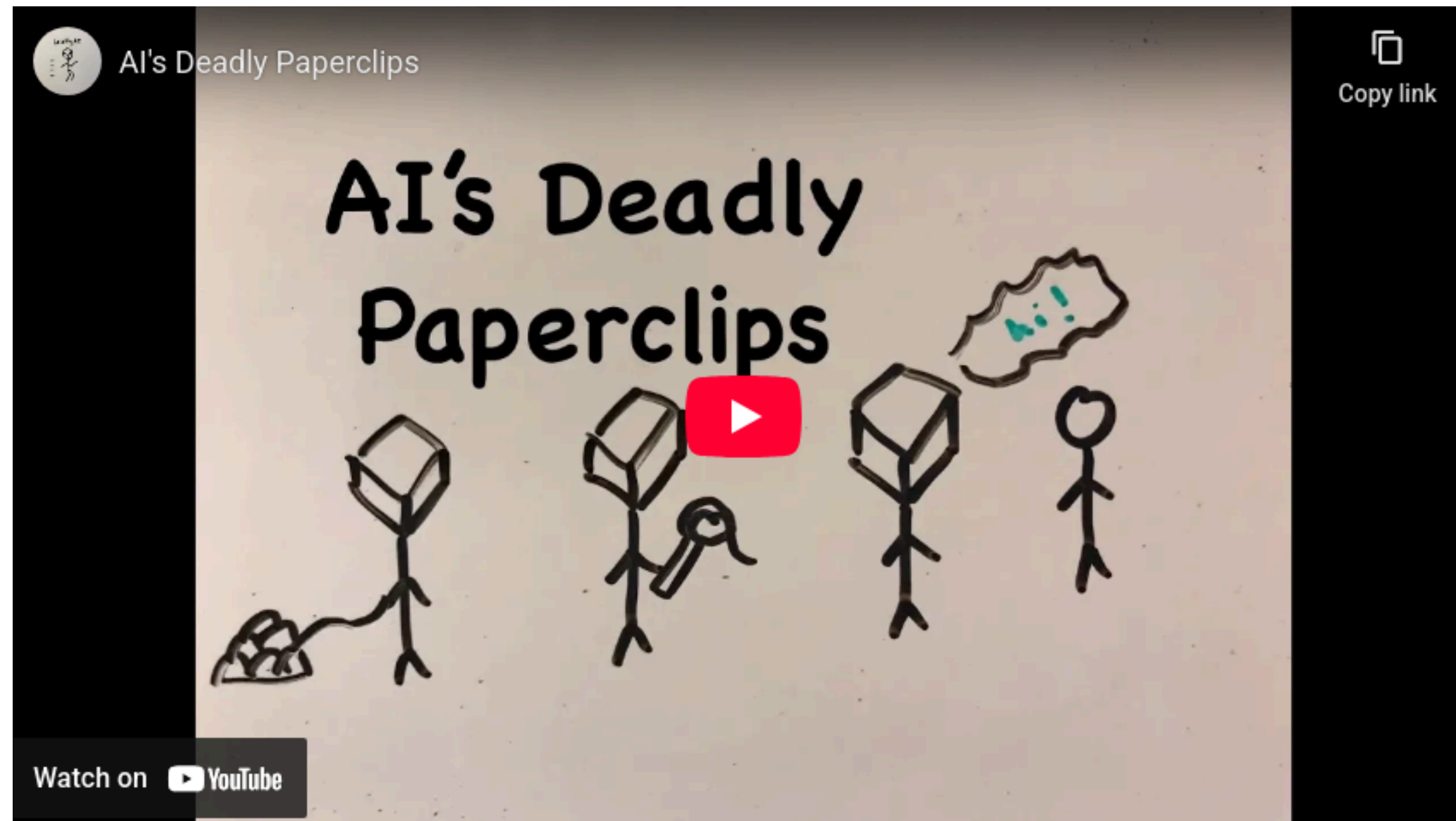
$\text{Risk} = \text{Likelihood} \times \text{Impact}$

Likelihood = probability of failure

Impact = severity of harm

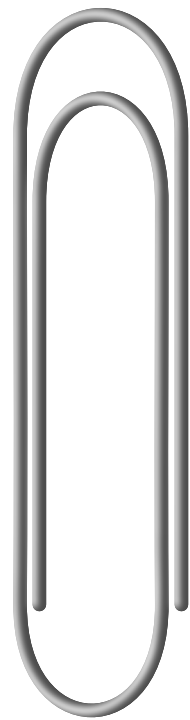
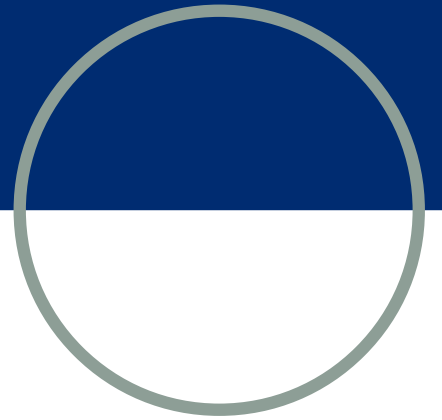


# WHERE DOES RISK IN AI COME FROM?





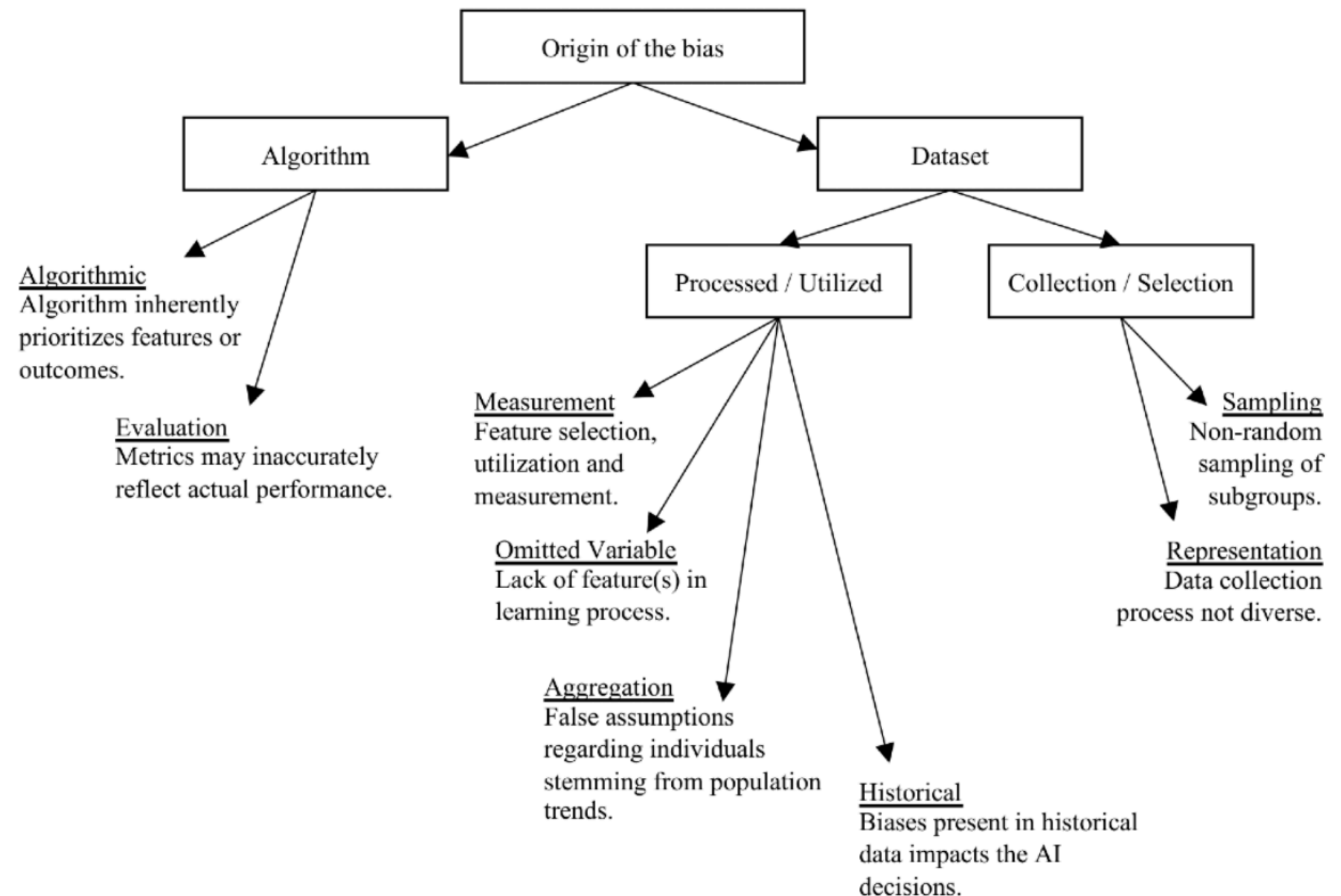
# WHERE DOES RISK IN AI COME FROM: MISALIGNED OBJECTIVES



# WHERE DOES RISK IN AI COME FROM: BIASED DATA

## Bias in data

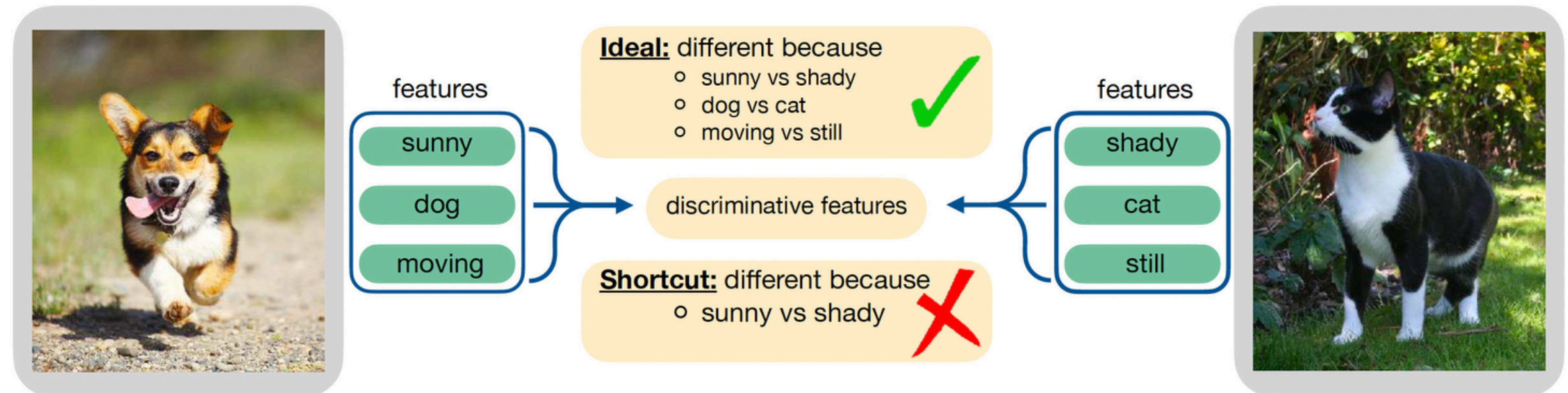
Mehrabi, Ninareh, et al.  
"A survey on bias and fairness in machine learning." ACM computing surveys (CSUR) 54.6 (2021): 1-35.



# WHERE DOES RISK IN AI COME FROM: MODEL

**Model**

Overfitting  
Shortcut learning



# WHERE DOES RISK IN AI COME FROM: HUMAN USE

Humans

Overreliance on AI  
Poor interpretability  
Misuse  
Automation bias





# WHERE DOES RISK IN AI COME FROM: CONTEXT

## Adversarial Machine Learning

### Context

Domain Shift  
Adversarial Behavior  
Emergent Behavior





**What risks can you foresee  
AI having in these domains  
(1 each):  
1) Healthcare 2) Defense  
3) Autonomus Vehicles**

**Talk to your neighbor and discuss for 10  
minutes**

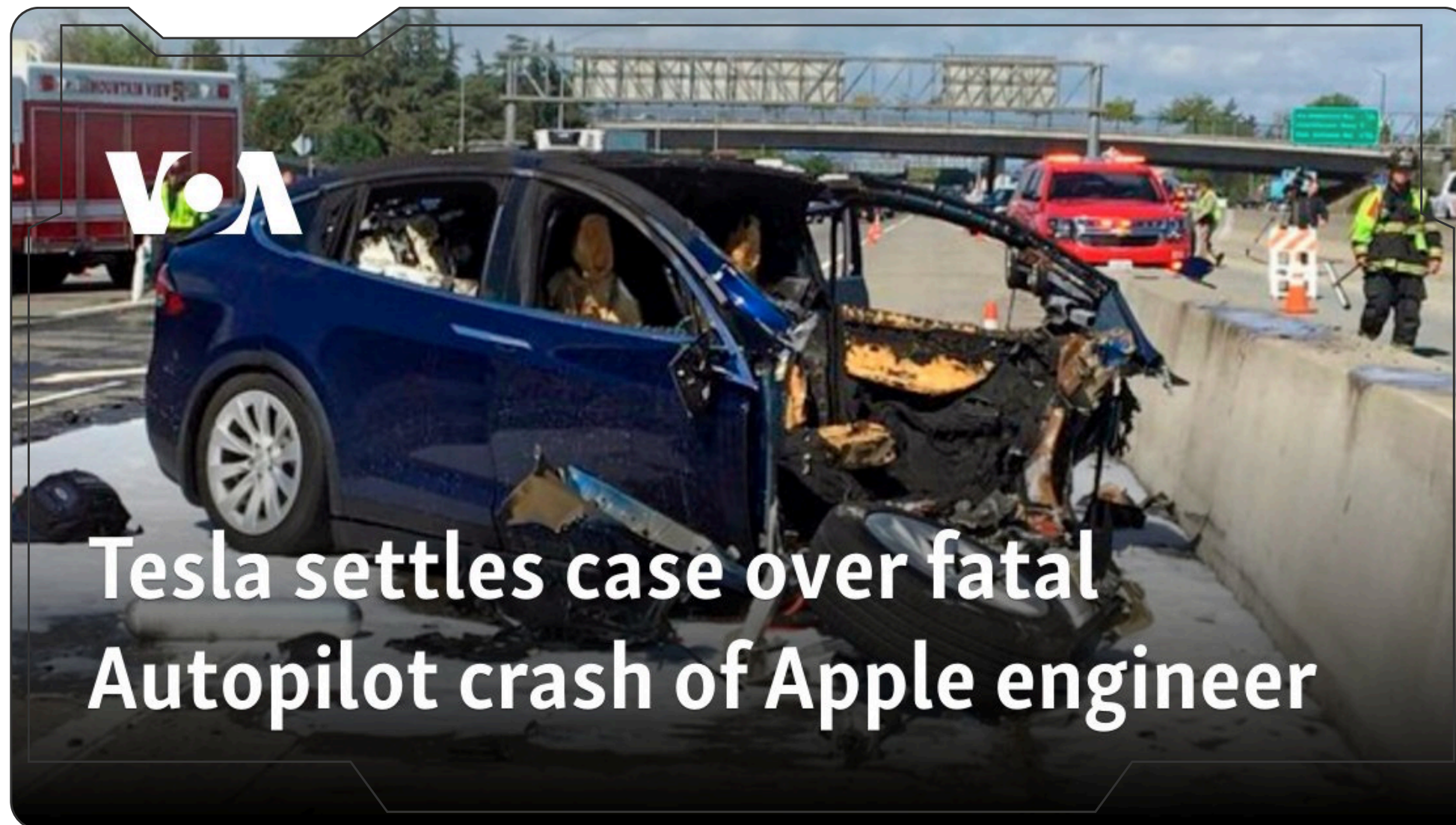


# HOW CAN WE MINIMIZE RISK?

Learn from failures!  
(postmortem analysis)



# AI FORENSICS



AI Forensics is the study of AI Incidents, aiming to understand the harm they have caused, the spread of the impact, and the root cause of the issue.

- \* **AI Incident**  
Something that “goes wrong” and inadvertently causes harm to humans.
- \* **Postmortem Analysis of AI Incidents**  
Looking at the incident from the “after the fact” perspective and performing an investigation.
- \* **AI Incident Database**  
An aggregated forum of AI Incidents, collecting and noted by the community at large.

# FORENSICS

## \* Collect Evidence

What pieces of information are relevant to what transpired? In the case of a home invasion, what was the entry point? What items were taken? What was the murder weapon?

## \* Analyze Evidence

Are there fingerprints on the door or murder weapon ? Any blood stains?

## \* Create a Recount of Events Transpired



Forensics, as we have all seen in murder mysteries, aims to help investigators better understand what transpired, why it happened, and how it was achieved.

# FORENSIC PROCEDURE

## \* What devices were involved?

Is this solely software and lives on the cloud? Is it integrated into hardware?

## \* Analyze Evidence

Are there fingerprints on the door or murder weapon ? Any blood stains?

## \* Create a Recount of Events Transpired

### Forensic Procedure

In subject area: Computer Science

Forensic Procedure is defined as a structured process involving steps like acquisition, analysis, evaluation, and presentation of digital evidence in a systematic and legal manner to investigate incidents related to digital devices.

AI generated definition based on: [Digital Forensics Processing and Procedures, 2013](#)

Of course, unless you are Shawn and Gus, you can't just break the rules when conducting forensic analysis. You have to follow a strict standard operating procedure. At a high-level, you have three key steps.



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**Search the AI Incident Database. Find the worst harm that has come from AI integration.**

**Talk to your neighbor and discuss for 10 minutes**



# Understanding the Types of Harm AI can Introduce

Subtitle





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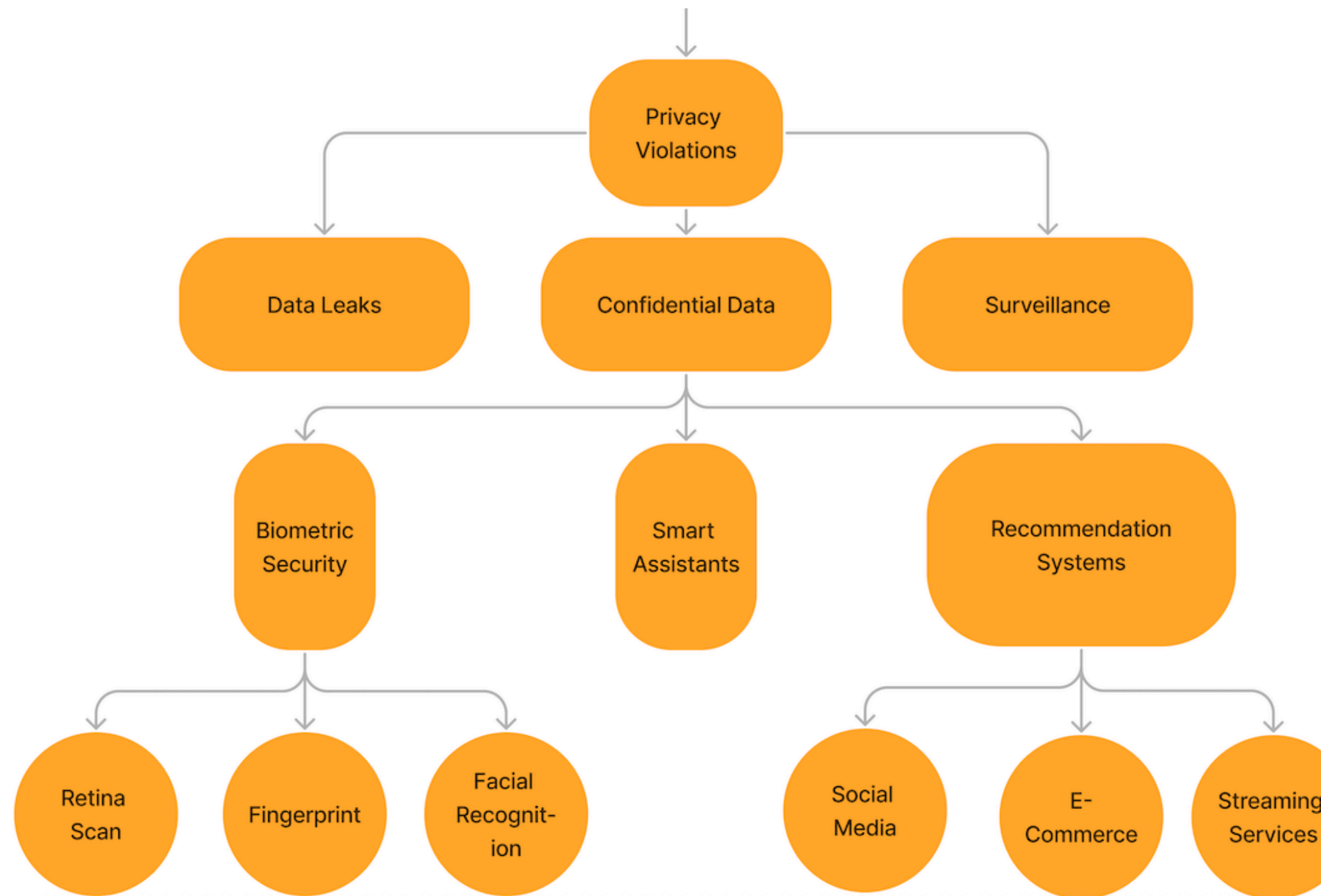
**Why is it important to  
categorize harm?**



# Why is it important to categorize harm?

- \* Group Incidents
- \* Identify Trends
- \* Plan Future Mitigation Strategies

# WHAT HARM MATERIALIZED?



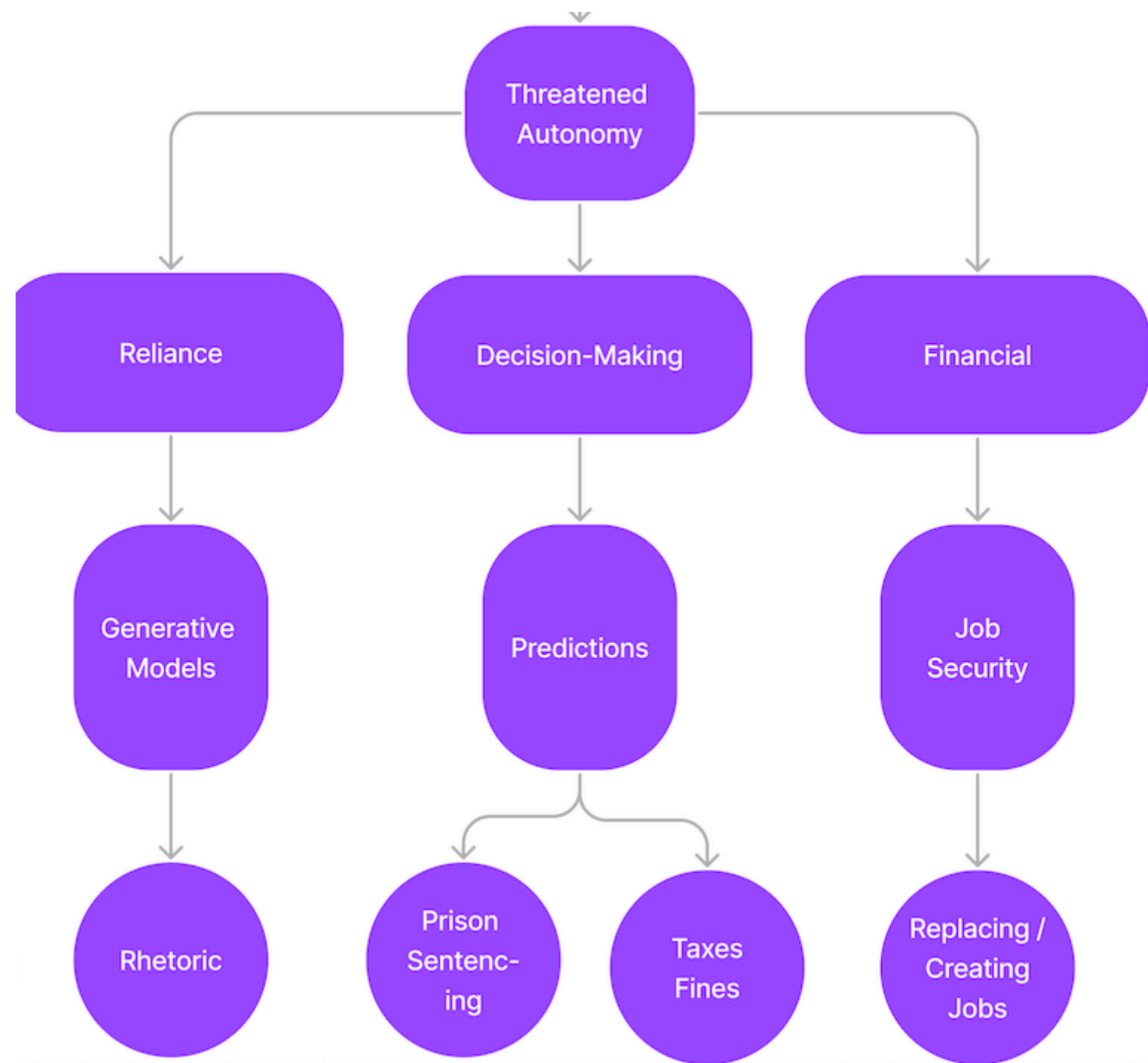
## \* Privacy Violations

How was human privacy violated? Are long terms of agreements, public spaces subject to facial recognition ethical?

## \* Threatened Autonomy

## \* Influenced Well-Being

# WHAT HARM MATERIALIZED?



## \* Privacy Violations

How was human privacy violated? Are long terms of agreements, public spaces subject to facial recognition ethical?

## \* Threatened Autonomy

We've worked hard to make machines autonomous, but what about humans? The old saying, you are who you hang around.

## \* Influenced Well-Being



# WHAT HARM MATERIALIZED?



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## \* Threatened Autonomy

We've worked hard to make machines autonomous, but what about humans? The old saying, you are who you hang around.

## \* Influenced Well-Being

What are the different ways that human well-being can be undermined?



**Identify one of the three types of harms and think about a potential mitigation strategy.**

**Talk to your neighbor and discuss for 5 minutes**

# RISK MANAGEMENT STRATEGIES

There has been a **LOT** of work in this area



## The Microsoft Responsible AI Standard

Explore guidance from Microsoft on how to design, build, and test AI systems.



PRESIDENTIAL ACTIONS

## ENSURING A NATIONAL POLICY FRAMEWORK FOR ARTIFICIAL INTELLIGENCE

Executive Orders | December 11, 2025



# ASILOMAR AI PRINCIPLES

## Asilomar AI Principles

The Asilomar AI Principles, coordinated by FLI and developed at the Beneficial AI 2017 conference, are one of the earliest and most influential sets of AI governance principles.

Signatures

**5720**

Add your  
signature

Published

11 August, 2017

One of the earliest and most influential sets of AI principles to date. I dare say that each subsequent set of principles today stems from these. These directly influence the mitigation strategies we employ today

# Ethics and Values

6) **Safety:** AI systems should be safe and secure throughout their operational lifetime, and verifiably so where applicable and feasible.

7) **Failure Transparency:** If an AI system causes harm, it should be possible to ascertain why.

8) **Judicial Transparency:** Any involvement by an autonomous system in judicial decision-making should provide a satisfactory explanation auditable by a competent human authority.

9) **Responsibility:** Designers and builders of advanced AI systems are stakeholders in the moral implications of their use, misuse, and actions, with a responsibility and opportunity to shape those implications.

10) **Value Alignment:** Highly autonomous AI systems should be designed so that their goals and behaviors can be assured to align with human values throughout their operation.

11) **Human Values:** AI systems should be designed and operated so as to be compatible with ideals of human dignity, rights, freedoms, and cultural diversity.

12) **Personal Privacy:** People should have the right to access, manage and control the data they generate, given AI systems' power to analyze and utilize that data.

13) **Liberty and Privacy:** The application of AI to personal data must not unreasonably curtail people's real or perceived liberty.

14) **Shared Benefit:** AI technologies should benefit and empower as many people as possible.

15) **Shared Prosperity:** The economic prosperity created by AI should be shared broadly, to benefit all of humanity.

16) **Human Control:** Humans should choose how and whether to delegate decisions to AI systems, to accomplish human-chosen objectives.

17) **Non-subversion:** The power conferred by control of highly advanced AI systems should respect and improve, rather than subvert, the social and civic processes on which the health of society depends.

18) **AI Arms Race:** An arms race in lethal autonomous weapons should be avoided.

# ASILOMAR AI PRINCIPLES

## Discussion:





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**Do you agree or disagree  
with the responsibility of  
AI designers and builders?**

**Talk to your neighbor and discuss for 5  
minutes**



# NIST RISK MANAGEMENT FRAMEWORK

## Categorize System

- identifies the information systems that need to be protected and assigns them to a security category based on the potential impact on the organization if they were to be compromised.



# NIST RISK MANAGEMENT FRAMEWORK

## Security Controls

- Based on the security category, the organization selects a set of security controls from NIST's Special Publication 800-53 to protect the information systems.

## CHAPTER THREE THE CONTROLS .....

|      |                                                                       |
|------|-----------------------------------------------------------------------|
| 3.1  | ACCESS CONTROL .....                                                  |
| 3.2  | AWARENESS AND TRAINING .....                                          |
| 3.3  | AUDIT AND ACCOUNTABILITY .....                                        |
| 3.4  | ASSESSMENT, AUTHORIZATION, AND MONITORING .....                       |
| 3.5  | CONFIGURATION MANAGEMENT .....                                        |
| 3.6  | CONTINGENCY PLANNING .....                                            |
| 3.7  | IDENTIFICATION AND AUTHENTICATION .....                               |
| 3.8  | INCIDENT RESPONSE .....                                               |
| 3.9  | MAINTENANCE .....                                                     |
| 3.10 | MEDIA PROTECTION .....                                                |
| 3.11 | PHYSICAL AND ENVIRONMENTAL PROTECTION .....                           |
| 3.12 | PLANNING .....                                                        |
| 3.13 | PROGRAM MANAGEMENT .....                                              |
| 3.14 | PERSONNEL SECURITY .....                                              |
| 3.15 | PERSONALLY IDENTIFIABLE INFORMATION PROCESSING AND TRANSPARENCY ..... |
| 3.16 | RISK ASSESSMENT .....                                                 |
| 3.17 | SYSTEM AND SERVICES ACQUISITION .....                                 |
| 3.18 | SYSTEM AND COMMUNICATIONS PROTECTION .....                            |
| 3.19 | SYSTEM AND INFORMATION INTEGRITY .....                                |
| 3.20 | SUPPLY CHAIN RISK MANAGEMENT .....                                    |

# NIST RISK MANAGEMENT FRAMEWORK

- Monitoring Security Controls
  - The organization continuously monitors the security controls to ensure that they remain effective in protecting the information system. Any changes to the system or the environment may require a reassessment.



# NIST AI RMF

## NIST RMF GOVERNANCE FUNNEL



### Govern Accountability

Embedding accountability throughout the process



### Map AI Systems

Understanding where AI is operating



### Measure Risks

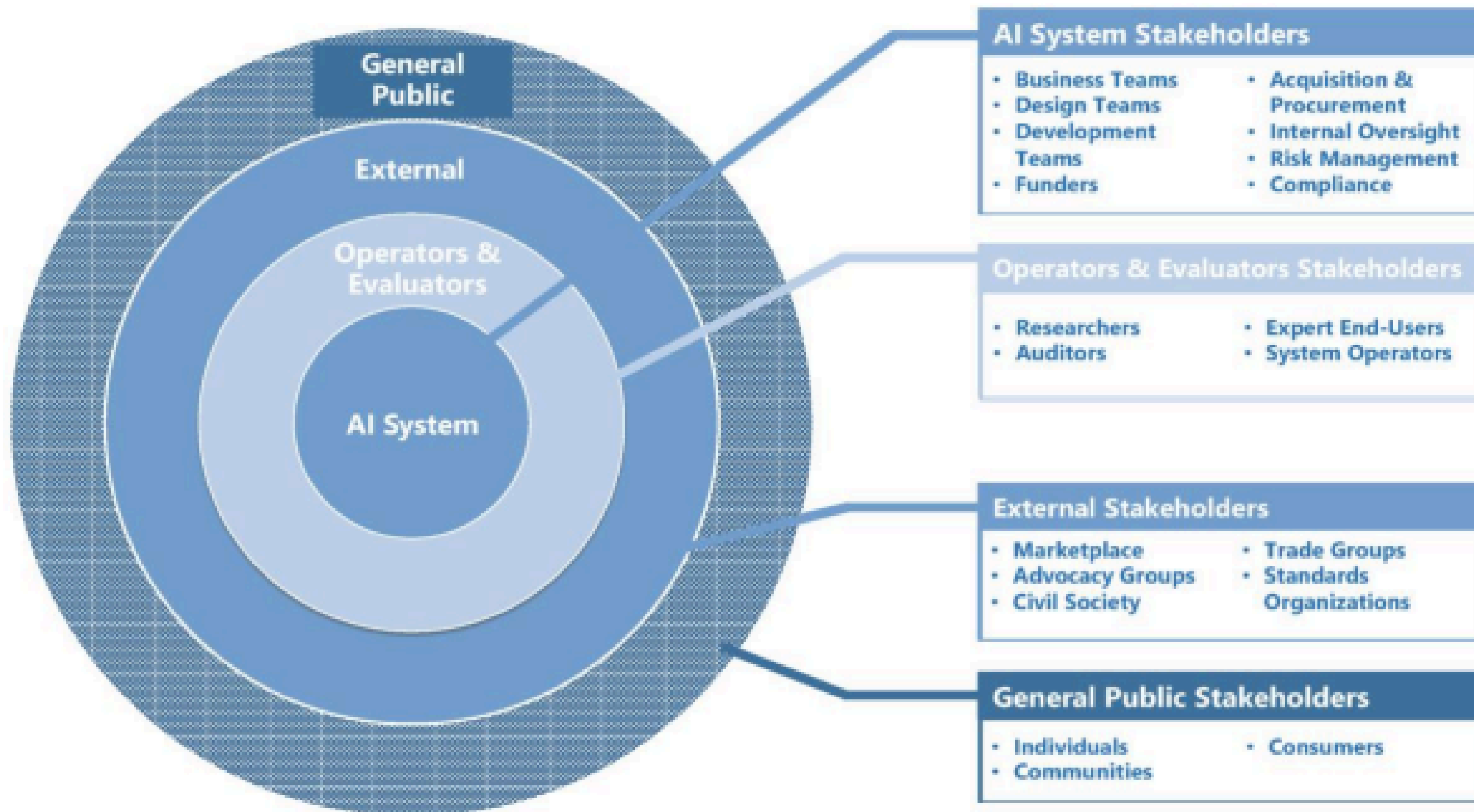
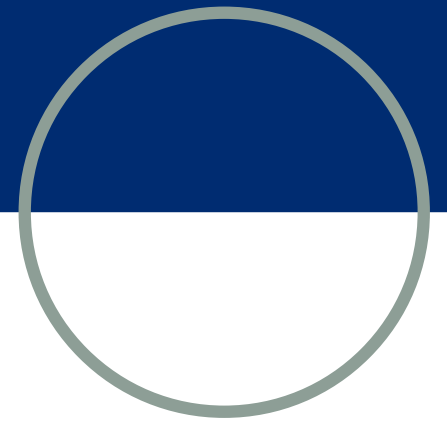
Assessing the nature and magnitude of risks



### Manage Controls

Implementing controls to mitigate risks

# NIST AI RMF: AUDIENCES





# NIST AI RMF

## Trustworthy AI: Risks & Characteristics

### Technical

- Accuracy
- Reliability
- Robustness
- Resilience or Security

### Socio Technical

- Explainability
- Interpretability
- Privacy
- Safety
- Managing Bias

### Guiding Principles

- Fairness
- Accountability
- Transparency



# NIST AI RMF



# MICROSOFT RESPONSIBLE AI PRINCIPLES

## Defining what's important

We've identified six principles that we believe should guide AI development and use.



### Fairness

AI systems should treat all people fairly.

How might an AI system allocate opportunities, resources, and information in ways that are fair to the humans who use it?

[See fairness in action](#)



### Reliability and safety

AI systems should perform reliably and safely.

How might a system function well for people across different use conditions and contexts, including ones that it wasn't originally intended for?

[See reliability and safety in action](#)



### Privacy and security

AI systems should be secure and respect privacy.

How might a system be designed to support privacy and security?

[See privacy and security in action](#)



### Inclusiveness

AI systems should empower everyone and engage all people, regardless of their backgrounds.

How might a system be designed to be inclusive for people of all abilities?

[See inclusiveness in action](#)



### Transparency

AI systems should be understandable.

How can we ensure that people correctly understand the capabilities of a system?

[See transparency in action](#)



### Accountability

People should be accountable for AI systems.

How can we create oversight so that humans can be accountable and in control?

[See accountability in action](#)



Ask Microsoft



# BUILDING ETHICAL AI

Your team has been invited to submit a bid to design an AI system to identify patients at risk of heart failure in Baltimore within a local healthcare system. Tabular data with patient demographics and biological information has been made available to you.

This is all the information you have and no further details are available.

What would be your strategy? Come up with it in your group and write it down and post it on Piazza.

Think through the following:

- What questions remain?
- What assumptions can you make?
- How would you uphold the AI principles we have discussed?
- What are some potential risks?
- How would you mitigate risk?

**Thank You!**





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UNIVERSITY